26900111 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECHEA01 - FORENSIC CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which statement about explosive classification is correct? [An] / [CO1]

   a). Each substance has only one possible classification label

   b). All industrial explosives are low explosives

   c). All mixtures are primary explosives

   d). One substance can be described using more than one classification scheme

2. Which cannabis product is specifically derived from the resin of the cannabis plant? [U] / [CO2]

   a). Charas

   b). Ganja

   c). Opium

   d). Cocaine

3. Toxicology is the study of: [K] / [CO3]

   a). Drugs only

   b). Poisons and their effects

c). Microorganisms d). Nutrition

4. The approximate pH range over which phenolphthalein exhibits its characteristic pink colour is: [K] / [CO4]

a). 1–3

b). 4–6

c). 8.2–10

d). above 13

5. Chain of custody is important because it: [U] / [CO2]

a). Changes the sample b). Establishes documented continuity of evidence c). Increases drug concentration d). Removes contaminants

6. Which of the following can cause an accidental fire? [An] / [CO1]

a). Defective electrical wiring b). Insurance fraud

c). Revenge d). Concealment of crime

7. Ethanol in alcoholic beverages is mainly produced by: [K] / [CO2]

a). Oxidation b). Fermentation

c). Hydrolysis d). Sublimation

8. Which poison commonly produces convulsions? [E] / [CO3]

a). Strychnine b). Ethanol

c). Morphine d). Aspirin

9. The most sensitive multi-element technique, also capable of isotope-ratio fingerprinting of heavy metals, is: [K] / [CO4]

a). AAS b). ICP-MS

c). UV-Vis d). TLC

10. Why should adhesive tape be examined as a layered material? [K] / [CO4]

a). Its adhesive contains no b). Its backing is always c). Its backing and adhesive d). Only its length can be

information	chemically identical	may provide different comparison features	compared

11. Fires involving cooking oils and fats are classified as: [A] / [CO1]

    a). Class A

    b). Class B

    c). Class D

    d). Class F

[1x11 = 11]

Part B

Short Answer Questions

Answer any **8** questions

Each question carries **3** marks

12. Define adulteration and give one general beverage example. [U] / [CO2]

13. Explain the role of forensic toxicology in determining the cause and manner of death. [A] / [CO3]

14. Classify dyes by chemical structure, with an example of each. [K] / [CO4]

15. Write a short note on chromatography in forensic toxicology. [An] / [CO3]

16. State the basic roles of gas chromatography and mass spectrometry in GC MS. [A] / [CO1]

17. What are synthetic drugs? [U] / [CO2]

18. Explain the procedure for collecting stomach contents during autopsy. [An] / [CO3]

19. Define Rf and explain its use in ink comparison. [K] / [CO4]

20. Mention four presumptive tests used in forensic drug analysis. [U] / [CO2]

21. What is secondary blast injury? [An] / [CO1]

[3x8 = 24]

Part C

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

22. Write a short account of cocaine and amphetamine type stimulants. [U] / [CO2]

23. Describe the emergency care measures provided to patients with acute poisoning. [A] / [CO3]

24. Explain addition and condensation polymerisation with mechanism and forensic examples. [K] / [CO4]

25. Explain the collection, preservation, and packing of fire debris. [An] / [CO1]

26. Explain sensitivity and stability of explosives. [An] / [CO1]

[5x3 = 15]

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

27. Explain how pharmacological, chemical and legal classifications complement one another. Use examples without treating legal categories as permanent or universal. [U] / [CO2]

28. Describe the general management of poisoning. Explain the emergency care of poisoned patients, including stabilization, decontamination, supportive care, and enhanced elimination of poisons. [K] / [CO3]

29. Compare composition based dating information with aging based measurements. Explain what each can contribute and why neither always provides a precise writing date. [K] / [CO4]

[10x2 = 20]